Introducing Zearn Summer 2025 Math Intensive Series

Zearn's Summer 2025 Intensive Series are designed based on learnings of the billions of math problems completed on our math platform: where students struggle and the specific foundational support that kids need to access grade-level math content.

During the academic year, we identify and recommend these foundational lessons to teachers so they can be used for deeper interventions. Our Summer Series pulls forward these recommendations into coherent summer math experiences. Each Summer 2025 Intensive Series offers coherent, focused four-to-six-week sequences that build the strong foundations all rising 1st through 9th graders need to move forward during the 2025-2026 academic year.

Comprehensive materials for 4-to-6-week programs

Each Series consists of top-rated materials that can be used flexibly across summer programs:

- **Daily digital math lessons** offer a consistent structure of learning activities, designed to accelerate math learning by integrating unfinished learning into the context of new learning. Students explore new concepts with real on-screen teachers, visualization of every math concept, interactive problem solving, and guided paper-and-pencil Student Notes.
- **Materials for hands-on problem solving** offer teachers and tutors daily application problems they can use to facilitate lively math discussions where students explore different ways to solve problems, unlocking creativity and joy with math.
- **Real-time reports on student learning** provide visibility into student learning and where students need additional support to move forward in their learning.

Focused on the big ideas of math

Each Summer Math Intensive Series deeply explores the essential math content that will help rising 1st–9th graders unlock grade-level math learning this fall. In younger grades, students explore one big idea, like addition and subtraction. In older grades, students tackle longer sequences covering a few big ideas, building a strong foundation in critical capstone content that they will need for success in later grades.

Recommended usage

In each 4-to-6-week program, **rising 1st through 9th graders** should complete 1 lesson each day (~30 minutes).

Explore the content of Zearn's Summer 2025 Math Intensive Series on the following pages.

Summer Math Intensive Series: Rising 1st Graders

For rising 1st graders, the Kindergarten Summer Intensive Series focuses on the essentials of addition and subtraction to set students up for success in 1st grade.

Mission	Content for Rising 1st Graders: 33 Lessons			
	Standards Addressed	**Mission Title**	**Lessons**	**Topics**
GKM4	K.NS.2a K.NS.2b K.NS.2c K.NS.1g K.CE.1a K.CE.1b K.CE.1c K.CE.1d K.CE.1e K.CE.1f	**Number Pairs, Addition and Subtraction to 10** This Mission marks the next exciting step in math for kindergartners—addition and subtraction! Students begin to harness their practiced counting abilities, knowledge of the value of numbers, and work with embedded numbers to reason about and solve addition and subtraction expressions and equations.	33	Topic A: Compositions and Decompositions of 2, 3, 4, and 5 Topic B: Decompositions of 6, 7, and 8 into Number Pairs Topic C: Addition with Totals of 6, 7, and 8 Topic D: Subtraction from Numbers to 8 Topic E: Decompositions of 9 and 10 into Number Pairs Topic F: Addition with Totals of 9 and 10 Topic G: Subtraction from 9 and 10 Topic H: Patterns with Adding 0 and 1 and Making 10

Summer Math Intensive Series: Rising 2nd Graders

For rising 2nd graders, the Grade 1 Summer Intensive Series focuses on counting, composing, and decomposing numbers. The Grade 1 Summer Intensive Series is intended to build a strong foundation of flexible strategies for all students to fall back on as they engage with Grade 2 content and beyond.

	Content for Rising 2nd Graders: 32 Lessons			
Mission	**VA SOL Addressed**	**Mission Title**	**Lessons**	**Topics**
G1M1	1.CE.1a 1.CE.1b 1.CE.1c 1.CE.1d 1.CE.1e 1.CE.1f 1.CE.1g 1.CE.1i 1.CE.1j 1.CE.1k 1.CE.1l	**Add and Subtract Small Numbers** In this Mission, students make significant progress towards fluency with addition and subtraction of numbers to 10 as they are presented with opportunities intended to advance them from counting all to counting on, which leads many students then to decomposing and composing addends and total amounts. In Kindergarten, students achieved fluency with addition and subtraction facts to 5. This means they can decompose 5 into 4 and 1, 3 and 2, and 5 and 0. They can do this without counting all. They perceive the 3 and 2 embedded within the 5	32	Topic A: Embedded Numbers and Decompositions Topic B: Counting On from Embedded Numbers Topic C: Addition Word Problems Topic D: Strategies for Counting On Topic E: The Commutative Property of Addition and the Equal Sign Topic F: Development of Addition Fluency Within 10 Topic G: Subtraction as an Unknown Addend Problem Topic H: Subtraction Word Problems Topic I: Decomposition Strategies for Subtraction Topic J: Development of Subtraction Fluency Within 10

Summer Math Intensive Series: Rising 3rd Graders

For rising 3rd graders, the Grade 2 Summer Intensive Series focuses on the capstone content of K–2: a deep understanding of the base ten system and flexibly being able to add and subtract. This is content that 3rd-grade standards assume students can access.

	Content for Rising 3rd Graders: 29 Lessons			
Mission	**VA SOL Addressed**	**Mission Title**	**Lessons**	**Topics**
G2M4	2.CE.1b 2.CE.1c	**Add, Subtract, and Solve** Equipped with a solid understanding of base ten, students will dive into decomposing and composing in addition and subtraction for numbers up to 200.	29	Topic A: Sums and Differences within 100 Topic B: Strategies for Composing a Ten Topic C: Strategies for Decomposing a Ten Topic D: Strategies for Composing Tens and Hundreds Topic E: Strategies for Decomposing Tens and Hundreds Topic F: Student Explanations of Written Methods

Summer Math Intensive Series: Rising 4th Graders

For rising 4th graders, the Grade 3 Summer Intensive Series focuses on multiplication and division, which is key to their success in later grades.

Mission	Content for Rising 4th Graders: 37 Lessons			
	VA SOL Addressed	**Mission Title**	**Lessons**	**Topics**
G3M3	3.CE.2a 3.CE.2b 3.CE.2c 3.CE.2d 3.CE.2e 3.CE.2f 3.PFA.1a 3.PFA.1b 3.PFA.1c 3.PFA.1d 3.PFA.1e	**Multiply and Divide Tricky Numbers** This Mission extends multiplication and division to all factors between 0 and 10. Students work deeply with the commutative, distributive, and associative properties and have problem-solving opportunities at the close of each topic.	21	Topic A: The Properties of Multiplication and Division Topic B: Multiplication and Division Using Units of 6 and 7 Topic C: Multiplication and Division Using Units up to 8 Topic D: Multiplication and Division Using Units of 9 Topic E: Analysis of Patterns and Problem Solving Including Units of 0 and 1 Topic F: Multiplication of Single-Digit Factors and Multiples of 10
G3M4	3.MG.2a	**Find the Area** In this Mission, students explore area as an attribute of two-dimensional figures and relate it to their prior understandings of multiplication.	16	Topic A: Foundations for Understanding Area Topic B: Concepts of Area Measurement Topic C: Arithmetic Properties Using Area Models Topic D: Applications of Area Using Side Lengths of Figures

Summer Math Intensive Series: Rising 5th Graders

For rising 5th graders, the Grade 4 Summer Intensive Series focuses on fractions and decimals. Students evaluate equivalence and learn to extend that understanding to decimal operations. The Summer Intensive Series is intended to ensure students have a solid grasp of these ideas to close out 4th grade successfully.

	Content for Rising 5th Graders: 53 Lessons			
Mission	**VA SOL Addressed**	**Mission Title**	**Lessons**	**Topics**
G4M5	4.CE.2h 4.CE.2j 4.CE.2k 4.PFA.1 4.NS.3e 4.NS.3a 4.NS.3b 4.NS.3c 4.NS.3d 4.CE.3a 4.CE.3b 4.CE.3c 4.MG.1 4.PS.1c 4.PS.1d	**Equivalent Fractions** This Mission teaches students how to manipulate fractions. Students compare fractions, evaluate equivalence, and learn that the same methods they used for whole number operations can be used to add, subtract, and multiply fractions.	38	Topic A: Decomposition and Fraction Equivalence Topic B: Fraction Equivalence Using Multiplication and Division Topic C: Fraction Comparison Topic D: Fraction Addition and Subtraction Topic E: Extending Fraction Equivalence to Fractions Greater Than 1 Topic F: Addition and Subtraction of Fractions by Decomposition Topic G: Repeated Addition of Fractions as Multiplication Topic H: Exploring a Fraction Pattern
G4M6	4.NS.3e 4.NS.5a 4.NS.5b 4.NS.5c 4.NS.4e 4.MG.1	**Decimal Fractions** Students learn a new and special notation for fractions in this Mission: decimals! They extend their understanding of the base ten system by first exploring equivalence between fractions and decimals. This understanding extends to comparing decimals and adding money.	15	Topic A: Exploration of Tenths Topic B: Tenths and Hundredths Topic C: Decimal Comparison Topic D: Addition with Tenths and Hundredths Topic E: Money Amounts as Decimal Numbers

Summer Math Intensive Series: Rising 6th Graders

For rising 6th graders, the Grade 5 Summer Intensive Series focuses on essential base ten and fraction operations work. The Summer Intensive Series is intended to ensure students are prepared for 6th-grade math as an extension of elementary school work.

	Content for Rising 6th Graders: 45 Lessons			
Mission	**VA SOL Addressed**	**Mission Title**	**Lessons**	**Topics**
G5M2	5.CE.4a 5.CE.4b 5.CE.1b 5.CE.3a 5.CE.3b 5.CE.3c 5.CE.3d 5.CE.3e 5.MG.1c	**Base Ten Operations** In this Mission, students will multiply multi-digit whole numbers in the first part of the Mission and divide them in the end. Each operation concludes with application work and measurement word problems.	29	Topic A: Mental Strategies for Multi-Digit Whole Number Multiplication Topic B: The Standard Algorithm for Multi- Digit Whole Number Multiplication Topic C: Decimal Multi-Digit Multiplication Topic D: Measurement Word Problems with Whole Number and Decimal Multiplication Topic E: Mental Strategies for Multi-Digit Whole Number Division Topic F: Partial Quotients and Multi-Digit Whole Number Division Topic G: Partial Quotients and Multi-Digit Decimal Division Topic H: Partial Quotients and Multi-Digit Decimal Division
G5M3	5.CE.2a 5.CE.2b 5.CE.2c	**Add and Subtract Fractions** In this Mission, students will develop flexibility with addition and subtraction of fractions so they can mentally or numerically solve, reason, and estimate their calculations. The Mission begins with concrete and pictorial work (using area models and number lines) and moves to numeric work with word problems.	16	Topic A: Equivalent Fractions Topic B: Making Like Units Pictorially Topic C: Making Like Units Numerically Topic D: Further Applications

Summer Math Intensive Series: Rising 7th Graders

For rising 7th graders, the Grade 6 Summer Intensive Series develops understanding of ratios and rates so that students can represent and think about them in multiple, flexible ways. This is important because these concepts will be the foundations of proportional relationships and linear equations.

	Content for Rising 7th Graders: 34 Lessons			
Mission	**VA SOL Addressed**	**Mission Title**	**Lessons**	**Topics**
G6M2	6.PFA.1a 6.PFA.1b 6.PFA.1c 6.PFA.2a 6.NS.1a 6.NS.1b 6.NS.1c 6.NS.1d 6.PFA.1e 6.PFA.1f 6.PFA.2b	**Introducing Ratios** In this Mission, students learn to understand and use the terms “ratio,” “rate,” “equivalent ratios,” “per,” “at this rate,” “constant speed,” and “constant rate,” and to recognize when two ratios are or are not equivalent. They represent ratios as expressions, and represent equivalent ratios with double number line diagrams, tape diagrams, and tables. They use these terms and representations in reasoning about situations involving color mixtures, recipes, unit pricing, and constant speed.	17	Topic A: What Are Ratios? Topic B: Equivalent Ratios Topic C: Representing Equivalent Ratios Topic D: Solving Ratio and Rate Problems Topic E: Part-Part-Whole Ratios Topic F: Let’s Put It to Work
G6M3	6.PFA.2a 6.NS.1a 6.NS.1b 6.NS.1c 6.NS.1d 6.PFA.1e 6.PFA.1f 6.PFA.2b	**Unit Rates and Percentages** In this Mission, students extend their developing understanding of ratios to dig deeper into ratio relationships and learn about rates and unit rates. Students are introduced to the idea of a percentage, which allows them to solve a variety of real-world problems.	17	Topic A: Making Sense of Division Topic B: Meanings of Fraction Division Topic C: Algorithm for Fraction Division Topic D: Fractions in Lengths, Areas, and Volumes Topic E: Let’s Put It to Work

Summer Math Intensive Series: Rising 8th Graders

For rising 8th graders, the Summer Intensive Series focuses on two big ideas introduced in Grade 7: arithmetic with signed numbers and solving multi-step equations. Students' work with rational number arithmetic and solving multi-step equations is woven throughout 8th grade, requiring students to have a certain degree of fluency with each to allow their working memory to focus on the new learning of Grade 8.

Mission	Content for Rising 8th Graders: 37 Lessons			
	VA SOL Addressed	**Mission Title**	**Lessons**	**Topics**
G7M5	7.CE.2a 7.CE.1a 7.PFA.3a 7.PFA.3b 7.PFA.3c 7.PFA.3d 7.PFA.3f 7.PFA.4a 7.PFA.4d 7.PFA.4f 7.PFA.4g	**Rational Number Arithmetic** In this Mission, students use tables and number line diagrams to represent sums and differences of signed numbers or changes in quantities represented by signed numbers such as temperature or elevation, becoming more fluent in writing different numerical addition and subtraction equations that express the same relationship.	15	Topic A: Interpreting Negative Numbers Topic B: Adding and Subtracting Rational Numbers Topic C: Multiplying and Dividing Rational Numbers Topic D: Four Operations with Rational Numbers Topic E: Solving Equations Where There are Negative Numbers Topic F: Let's Put It to Work
G7M6	7.CE.1a 7.PFA.3a 7.PFA.3b 7.PFA.3c 7.PFA.3d 7.PFA.3f 7.PFA.4a 7.PFA.4b 7.PFA.4c 7.PFA.4d 7.PFA.4f 7.PFA.4g	**Expressions, Equations, and Inequalities** In this Mission, Students learn algebraic methods for solving equations. Students solve linear inequalities in one variable and represent their solutions on the number line. They understand and use the terms "less than or equal to" and "greater than or equal to," and the corresponding symbols.	22	Topic A: Representing Situations of the Form $px + q = r$ and $p(x + q) = r$ Topic B: Solving Equations of the Form $px + q = r$ and $p(x + q) = r$ and Problems That Lead to Those Equations Topic C: Inequalities Topic D: Writing Equivalent Expressions

Summer Math Intensive Series: Rising 9th Graders

For rising 9th graders, the Summer Intensive Series deepens students' understanding of linear functions and associations in data, both essential concepts for a smooth transition into Algebra I. Algebra I explores how different types of functions behave, and linear functions serve as the launch point. A strong grasp of these concepts ensures students are prepared for the challenges ahead in high school math.

Mission	Content for Rising 9th Graders: 34 Lessons			
	VA SOL Addressed	**Mission Title**	**Lessons**	**Topics**
G8M5	8.PFA.2a 8.PFA.3b 8.PFA.3e 8.PFA.3c 8.PFA.3d 8.MG.2b 8.MG.2c 8.MG.2d	**Functions and Volume** In this Mission, students are introduced to the concept of a function. They describe functions as increasing or decreasing between specific numerical inputs and consider the inputs of a function to be values of its independent variable and its outputs to be values of its dependent variable.	21	Topic A: Inputs and Outputs Topic B: Representing and Interpreting Functions Topic C: Linear Functions and Rates of Change Topic D: Cylinders and Cones Topic E: Dimensions and Shapes
G8M6	8.PS.3c 8.PS.3d 8.PS.3e 8.PS.3f	**Associations in Data** In this Mission, students generate and work with bivariate data sets that has more variability than in previous Missions. Students describe scatter plots, using a term previously used to describe univariate data "cluster," and the new term "outlier." They fit lines to scatter plots and informally assess their goodness of fit by judging the closeness of the data points to the lines, and compare predicted and actual values. Students learn to understand and use the terms "two-way table," "bar graph," and "segmented bar graph," using two-way tables to investigate categorical data.	13	Topic A: Does This Predict That? Topic B: Associations in Numerical Data Topic C: Associations in Categorical Data Topic D: Let's Put It to Work